Student ID: 24IT087

Chandubhai S Patel Institute of Technology

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
B. TECH (IT/CE/CSE/AIML)

DEPSTAR

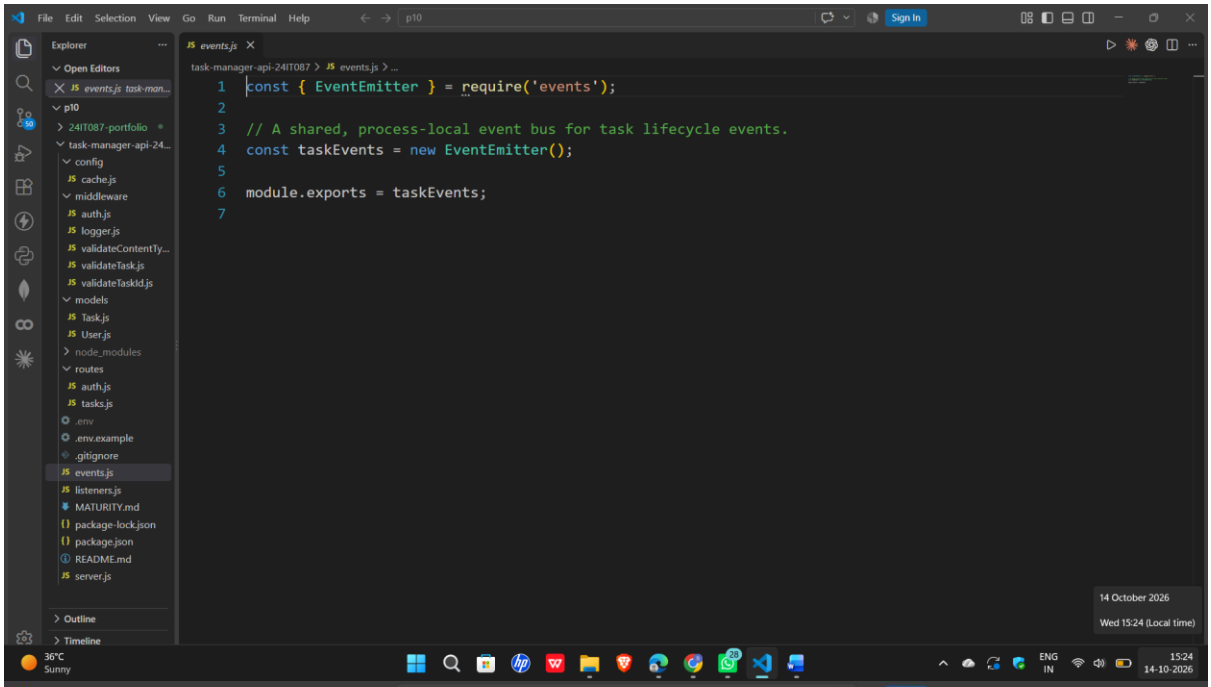
Subject: Advanced Web Development Frameworks (ITUE301) | Semester: 5th

Practical Statement:

- Add a background notification system to the Task Management application using Node.js built-in EventEmitter.
- When a new task is created via the API, emit a task-created event handled asynchronously.
- The event handler must log a notification (task title, timestamp, assigned user) without blocking the main response.
- The main API response must return immediately, before the event handler finishes executing.
- Demonstrate the difference between synchronous and asynchronous handling by logging timestamps at both the API response point and the event handler execution point.

Screenshots:

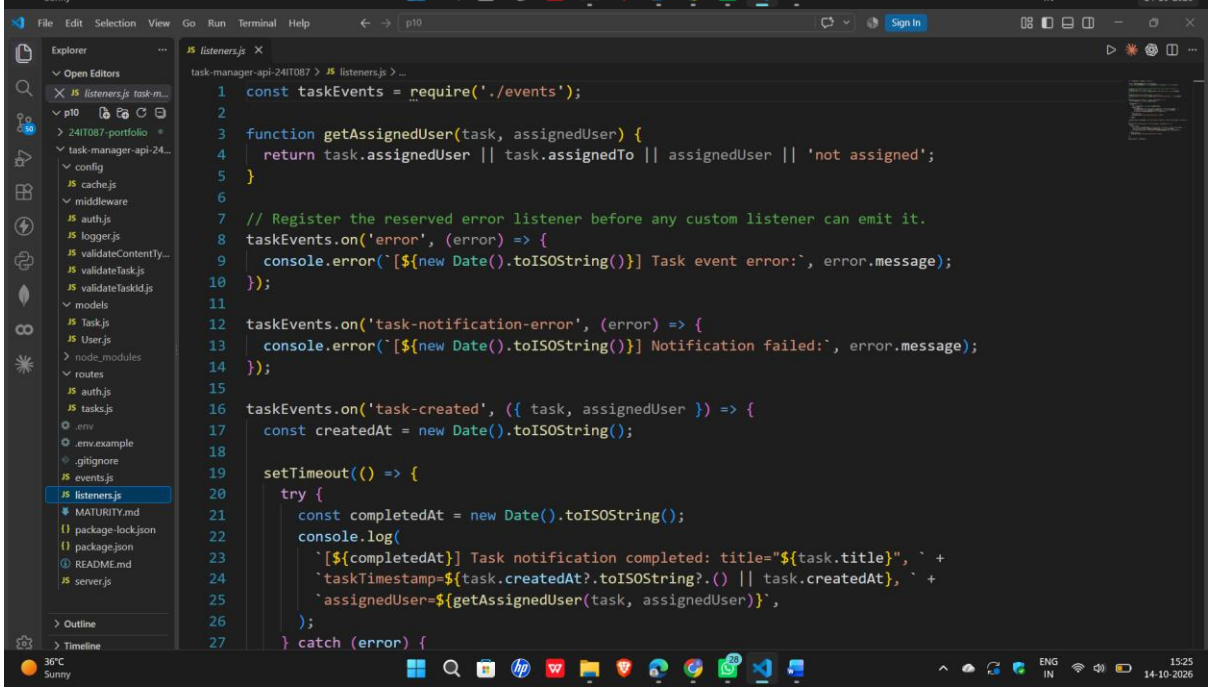
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```
// events.js
const { EventEmitter } = require('events');

// A shared, process-local event bus for task lifecycle events.
const taskEvents = new EventEmitter();

module.exports = taskEvents;
```



```
// listeners.js
const taskEvents = require('./events');

function getAssignedUser(task, assignedUser) {
  return task.assignedUser || task.assignedTo || assignedUser || 'not assigned';
}

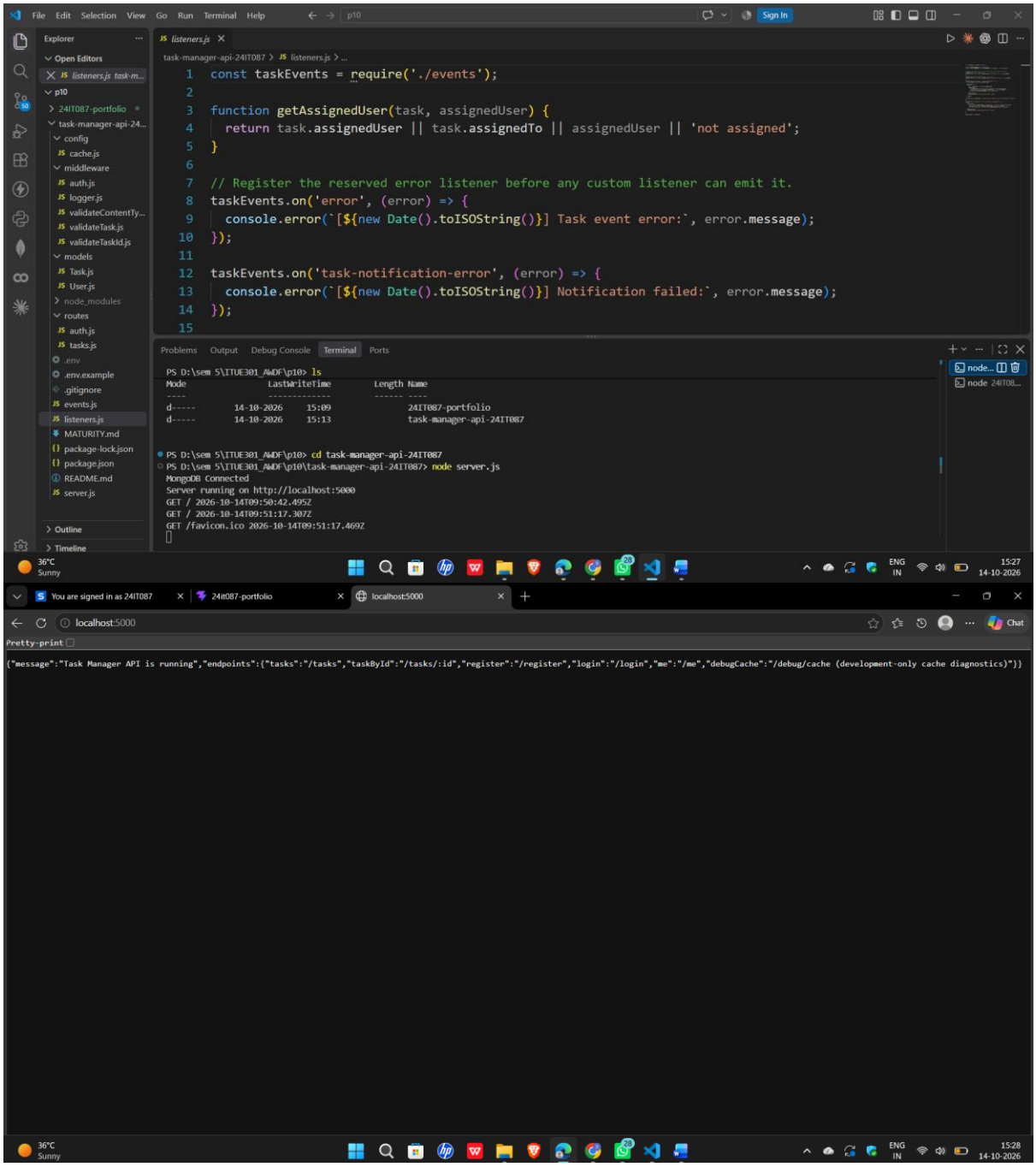
// Register the reserved error listener before any custom listener can emit it.
taskEvents.on('error', (error) => {
  console.error(`[${new Date().toISOString()}] Task event error:`, error.message);
});

taskEvents.on('task-notification-error', (error) => {
  console.error(`[${new Date().toISOString()}] Notification failed:`, error.message);
});

taskEvents.on('task-created', ({ task, assignedUser }) => {
  const createdAt = new Date().toISOString();

  setTimeout(() => {
    try {
      const completedAt = new Date().toISOString();
      console.log(
        `[${completedAt}] Task notification completed: title="${task.title}", ` +
        `taskTimestamp=${task.createdAt?.toISOString?.() || task.createdAt}, ` +
        `assignedUser=${getAssignedUser(task, assignedUser)}`,
      );
    } catch (error) {
```

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Github Repo Link:
https://github.com/digishasavaliya/ITUE301_AWDF.git